



Solar radiation management and its impact on Earth's hydrological cycle

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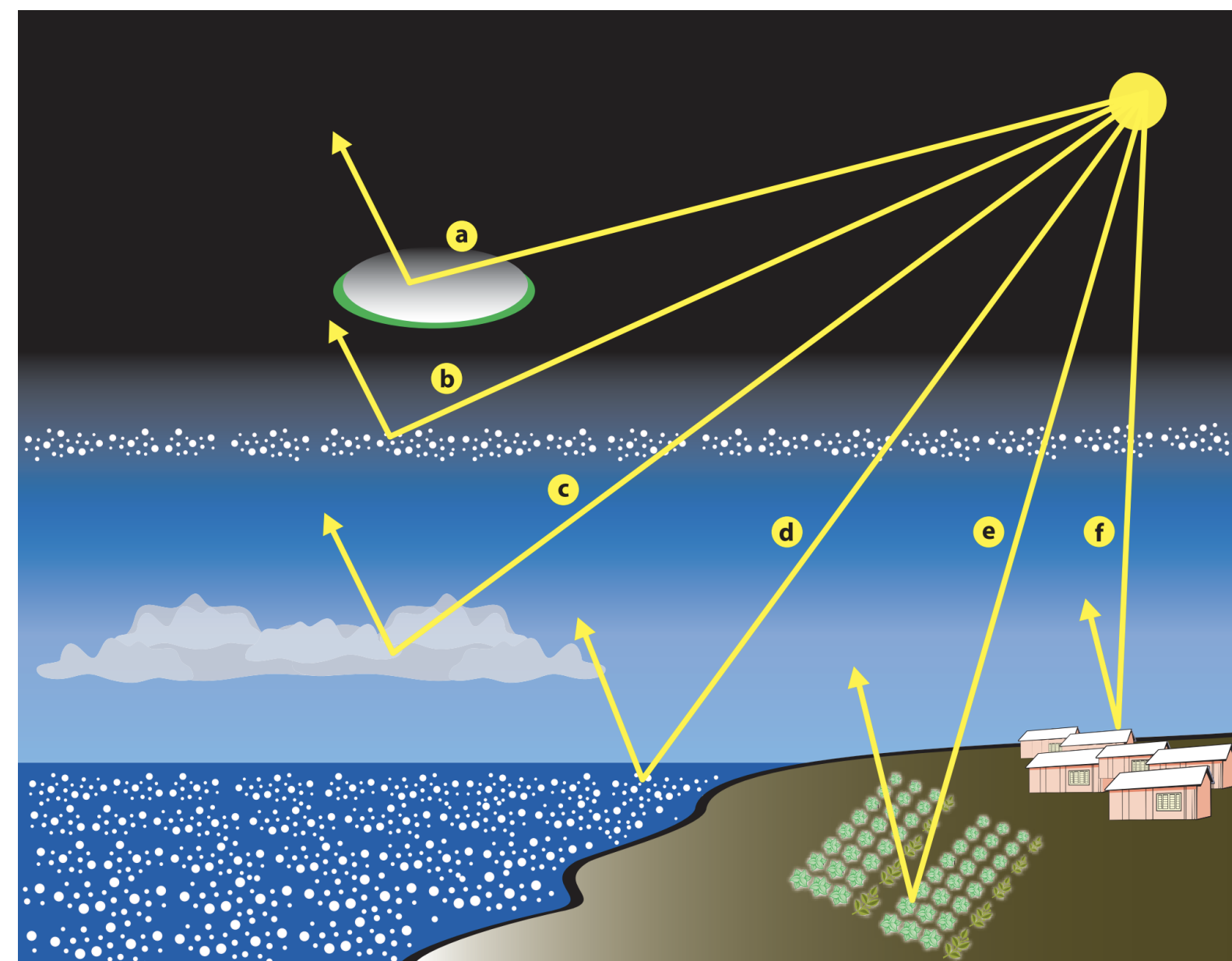
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INTRODUCTION

Increases in atmospheric CO₂ and other greenhouse gases exert a radiative forcing on the climate system by making it more difficult for heat to escape to space. Solar radiation management (SRM) or Solar Geoengineering is defined as the proposed artificial altering of this radiative balance in order to temporarily counteract some of the imbalance produced by the increase in atmospheric greenhouse gases (GHGs). This might be achieved in multiple ways, such as

- using satellites in space,
- injecting aerosols into the stratosphere
- brightening marine clouds
- making the ocean surface more reflective
- growing more reflective plants
- whitening roofs and other built structures

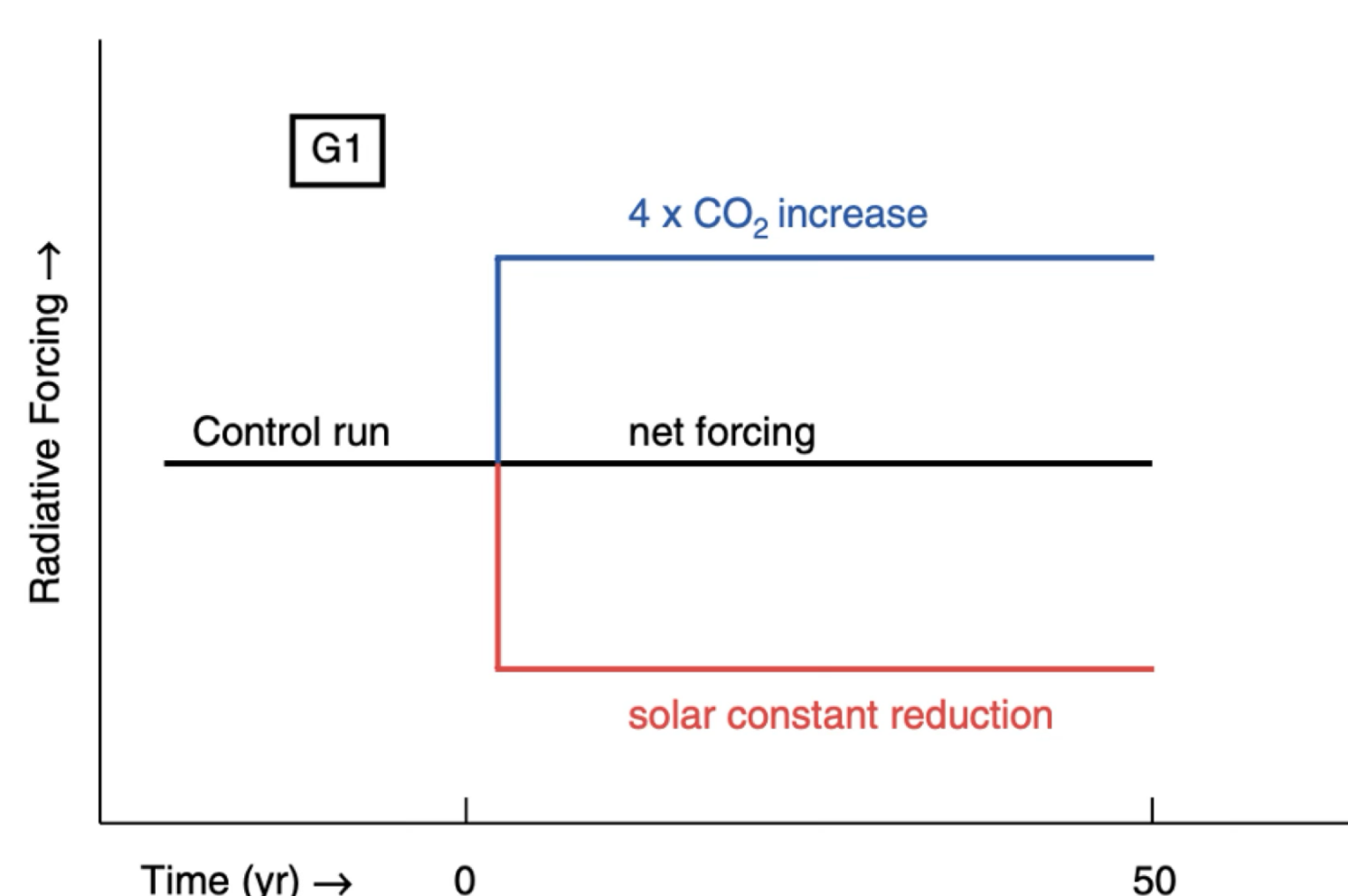


OBJECTIVES

- To understand the impact of SRM on Earth's hydrological cycle.
- To identify regions with the most pronounced benefits and risks under SRM interventions.
- To evaluate the implications of SRM on the Indian Summer Monsoon and South Asia's broader climate systems.

METHODOLOGY

Simulating such a technique in climate models is the main way to understand the possible impacts on the composition of the atmosphere and on the surface climate to determine its eventual feasibility, understand its possible impacts on ecosystems and populations and inform policymakers and stakeholders. The Geoengineering Model Intercomparison Project (GeoMIP) was proposed as a way to standardize SRM modeling experiments, allowing for a more robust comparison between model responses and determination of sources of uncertainties and areas for improvement.



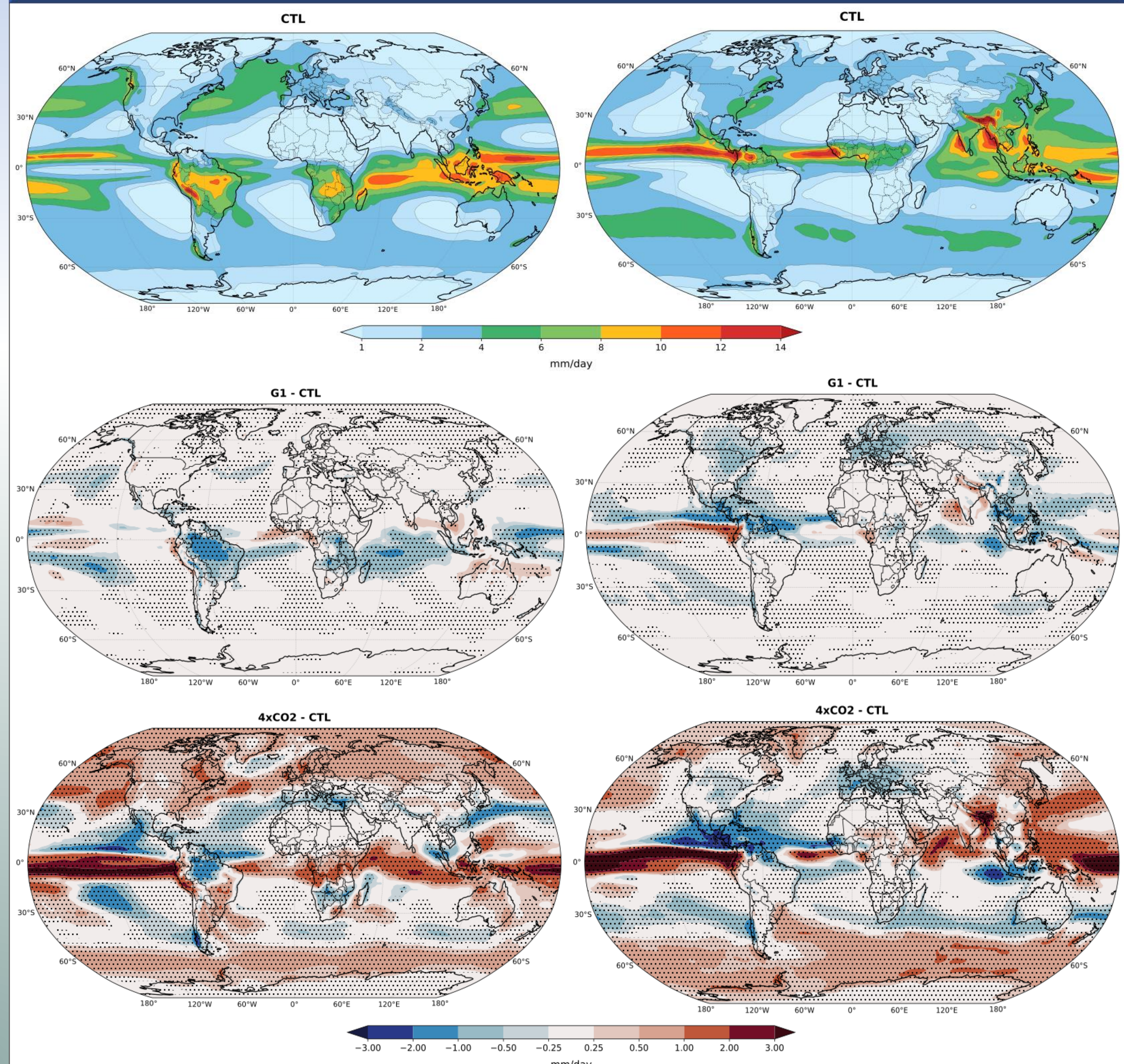
The G1 experiment assumes an idealized scenario in which a solar geoengineering scheme blocks enough solar radiation to counterbalance the warming that would occur if carbon dioxide concentrations were to quadruple. Source: Vioni et al. 2022.

The study employed multi model mean of eight GeoMIP5 models – BNU-ESM, CESM1-CAM5, CCSM4, GISS-E2R, HadGEM2-ES, IPSL-CM5A-LR, MIROC-ESM and MPI-ESM-LR, assessing global precipitation responses for 4xCO₂, G1 and, piControl (CTL) levels during boreal winter and summer.

REFERENCES

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RESULTS



Ensemble mean seasonal (DJF- left and JJA - right) precipitation differences between 4xCO₂ and G1 simulations w.r.t control climate. Stippling marks the regions of statistically significant differences (see the methodology section for the statistical measures)

- Vast stretches of NH mid-latitudes covering regions like Europe, Northern America, and Russia would witness reduction in precipitation in boreal summer in both the cases.
- Higher CO₂ levels lead to enhancement of precipitation in boreal winter with highest increase over Europe, Australia, and Northern America whereas, in boreal summer it leads to reduction in Australia, North America, South America, and Europe.
- Most parts of Asia, Antarctica, and Africa would witness consistent increase in precipitation in both the seasons, thus least affecting the seasonal contrast over these regions.
- Geoengineering of climate would lead to an overall drying of land as well as oceans in both the seasons.
- The drying of land is more severe over the continents in boreal summer season with Europe, North America, and Central America being the highest impacted regions.

CONCLUSION

- The geoengineered Earth will be very closer to the pre-industrial conditions and much better than the higher CO₂ warmer world.
- Regions like Europe, USA, and Russia, are prone to over drying during summer in a geoengineered world compared to near mitigated changes over other regions like South Asia, Africa, and Australia in both seasons.

Future Work

Building on this study, future research will focus on evaluating and understanding how different SRM strategies influence the following:

- Indian summer monsoon dynamics
 - Water deficit risk over major Indian river basins
 - Precipitation and temperature extremes in these basins
 - Global and as well South Asian streamflow and flood response.
- This regional-scale analysis will provide insights into the potential trade-offs and consequences of SRM on hydroclimatology in one of the most climate-vulnerable regions of the world.